BONUS · NEUROKIDS

Pattern Worksheets

Build logical thinking with fun pattern recognition and completion exercises

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Introduction: What Are Patterns?

> *A pattern is something that repeats in a predictable way. Once you find the rule, you can figure out what comes next!*

Patterns are everywhere around you. The stripes on a zebra, the tiles on a floor, the days of the week — they all follow a rule that repeats over and over. When you learn to spot patterns, you train your brain to think logically and solve problems.

Examples of Patterns

Colors: Red, Blue, Red, Blue, Red, Blue — the rule is "red then blue," repeating.

Shapes: Circle, Square, Circle, Square — two shapes taking turns.

Numbers: 2, 4, 6, 8 — each number grows by 2.

Sounds: Clap, Clap, Stomp, Clap, Clap, Stomp — a rhythm!

How to Solve a Pattern

Step 1: Say the pattern out loud. Hearing it helps your brain find the rhythm.
Step 2: Look at what repeats. Ask yourself: "Which part keeps coming back?"
Step 3: Guess what comes next. Use the rule you found to predict the next piece.
Step 4: Check your answer. Go back to the start and make sure your guess fits the rule all the way through.

Tip for Parents

Encourage your child to describe the pattern rule in their own words before filling in the blanks. This builds verbal reasoning alongside visual pattern recognition.

Important

There are no trick questions in these worksheets. Every pattern follows a clear, consistent rule. If a pattern seems confusing, go back to Step 1 and say it out loud.

SECTION 1

Color Patterns

Learn to recognize repeating, growing, and shrinking patterns using colors.

1.1 — AB Patterns

Each row follows an AB pattern (two colors taking turns). Fill in the blank cells to continue the pattern.

Row 1:



Row 2:



Row 3:



Row 4:



Row 5:



Row 6:



Key: R=Red, B=Blue, G=Green, Y=Yellow, P=Purple, O=Orange, Pk=Pink, W=White, Bk=Black, Br=Brown, T=Tan

1.2 — ABC Patterns

These patterns repeat three colors in order. Find the rule and fill in the blanks.

Row 1:

R B G R B G R ☐ ☐

Row 2:

Y Pk O Y Pk ☐ ☐ ☐

Row 3:

W Gy Bk W Gy ☐ ☐ ☐

Row 4:

P R B P R ☐ ☐ ☐

Row 5:

G Gd Sv G ☐ ☐ ☐

Key: R=Red, B=Blue, G=Green, Y=Yellow, Pk=Pink, O=Orange, W=White, Gy=Gray, Bk=Black, P=Purple, Gd=Gold, Sv=Silver

1.3 — AABB Patterns

In an AABB pattern each color appears twice before switching. Fill in the blanks.

Row 1:



Row 2:



Row 3:



Row 4:



Row 5:



Key: R=Red, B=Blue, G=Green, Y=Yellow, Pk=Pink, P=Purple, O=Orange, W=White, Bk=Black, Gy=Gray

1.4 — Growing Color Patterns

In these patterns, the number of items grows each time. Count carefully and fill in the blanks.

Row 1: The count goes up by 1 each time, alternating Red and Blue.



Write the answer: ___ Red

Row 2: Alternating Green and Yellow, with growing counts.



Write the answer: ___ Yellow

Row 3: Purple doubles each step.



Write the answer: ___ Purple (hint: 1, 2, 4, ...)

Row 4: Orange grows by 2 each step.



Write the answer: ___ Orange (hint: 1, 3, 5, ...)

1.5 — Shrinking Color Patterns

Now the counts go down! Figure out how much they shrink each time.

Row 1: Alternating Red and Blue, counts shrink by 1.



Write the answer: ___ Red

Row 2: Stars and Moons shrink by 1.



Write the answers: ___ Stars, ___ Moons

Row 3: Green shrinks by 2.



Write the answers: ___G, ___G

Row 4: Purple and Orange, shrinking by 1 each pair.



Write the answer: ___O

SECTION 2

Shape Patterns

Spot patterns using shapes, rotations, sizes, and combinations.

2.1 — Repeating Shape Sequences

Each row uses shape symbols. Find the repeating group and fill in the blanks.

Row 1: AB pattern



Row 2: ABC pattern



Row 3: AABB pattern



Row 4: ABBC pattern



Row 5: ABCD pattern



Row 6: ABAC pattern



Shapes: O = Circle, ▵ = Triangle, □ = Square, ☆ = Star, ◊ = Diamond

2.2 — Rotating Shapes

These patterns rotate. Arrows turn clockwise. Predict what direction comes next.

Row 1: Quarter turns clockwise



Row 2: Half turns



Row 3: Two clockwise quarter turns, then one back



Row 4: Counter-clockwise quarter turns



Row 5: Diagonal arrows rotating



Row 6: Repeating pair of rotations



2.3 — Size Sequences

These patterns change in size. S = Small, M = Medium, L = Large. Fill in the blanks.

Row 1: Growing cycle



Row 2: Shrinking cycle



Row 3: Grow and shrink



Row 4: Size with shape — circles grow then reset



Row 5: Two sizes alternating



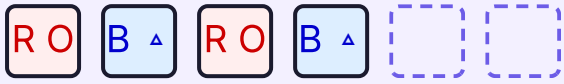
Row 6: Three sizes, each repeated



2.4 — Combined Shape-Color Patterns

Now the pattern uses BOTH a shape and a color. Pay attention to both!

Row 1:



Row 2:



Row 3: Color stays, shape changes



Row 4: Shape stays, color changes



Row 5: Both change independently



R=Red, B=Blue, G=Green, Y=Yellow, P=Purple. O=Circle, ▵=Triangle, □=Square, ☆=Star, ◊=Diamond

2.5 — Symmetry Patterns

Each row is a mirror pattern. The left side is given. Complete the right side so it mirrors the left.

Row 1: Mirror around the center line



Row 2:



Row 3: Mirror with colors



Row 4: Mirror with sizes



Row 5: Longer mirror



SECTION 3

Number Patterns

Discover patterns hidden in numbers — counting, growing, shrinking, and skipping.

3.1 — Counting by 2s, 5s, and 10s

Complete each counting sequence by filling in the blanks.

Count by 2:

2, 4, 6, 8, ___, ___, ___

12, 14, 16, ___, ___, ___

Count by 5:

5, 10, 15, 20, ___, ___, ___

35, 40, 45, ___, ___, ___

Count by 10:

10, 20, 30, ___, ___, ___

Count by 3:

3, 6, 9, 12, ___, ___, ___

Count by 4:

4, 8, 12, 16, ___, ___, ___

3.2 — Growing Number Sequences

Each sequence grows, but the amount it grows by might change! Look for the hidden rule.

Row 1: Add 1 more each time (+1, +2, +3, +4, ...)

1 2 4 7 11 ☐ ☐

Row 2: Triangular numbers (+2, +3, +4, +5, ...)

1 3 6 10 15 ☐ ☐

Row 3: Doubling

2 4 8 16 ☐ ☐

Row 4: Add 3, add 4, add 5, ...

5 8 12 17 ☐ ☐

Row 5: Square numbers

1 4 9 16 ☐ ☐

Row 6: Add 2, add 3, add 2, add 3, ...

1 3 6 8 11 ☐ ☐

3.3 — Decreasing Sequences

These numbers get smaller. Find the rule and fill in the blanks.

Row 1: Subtract 2

20 18 16 14 ☐ ☐

Row 2: Subtract 5

50 45 40 35 ☐ ☐

Row 3: Subtract 10

100 90 80 ☐ ☐ ☐

Row 4: Subtract 3

30 27 24 21 ☐ ☐

Row 5: Subtract 4

40 36 32 28 ☐ ☐

Row 6: Halving

64 32 16 8 ☐ ☐

3.4 — Skip Counting

Some numbers in the sequence are missing. Fill in every blank.

Row 1: Count by 3

3		9	12		18

Row 2: Count by 7

7	14		28		42

Row 3: Count by 6

6		18		30	

Row 4: Count by 8

8	16			40	48

Row 5: Count by 9

9		27		45	

Row 6: Count by 11

11		33	44		66

3.5 — Pattern Tables

Each table follows a rule that turns the Input into the Output. Find the rule, then fill in the missing outputs.

Table 1: Rule: multiply by 3

INPUT	OUTPUT
1	3
2	6
3	9
4	
5	

The rule is: Input × ___ = Output

Table 2: Find the rule

INPUT	OUTPUT
1	5
2	6
3	7
4	
5	

The rule is: Input + ___ = Output

Table 3: Find the rule

INPUT	OUTPUT
2	4

INPUT	OUTPUT
5	10
3	6
7	
10	

The rule is: Input × ___ = Output

SECTION 4

Challenge Patterns

Put your pattern skills to the test with matrices, multi-attribute puzzles, and creative challenges.

4.1 — Matrix Patterns

In each 3×3 grid, every row and every column follows a rule. Figure out what belongs in the cell marked "?".

Matrix 1: Each row uses 3 different shapes. Each column also uses 3 different shapes.

○	△	□
□	○	△
△	□	?

The missing shape is: ___

Matrix 2: Each row contains the numbers 1, 2, 3 in some order. Each column does too.

1	2	3
3	1	2
2	?	?

The missing numbers are: ___ and ___

Matrix 3: Each row has a Red, Blue, and Green cell. Each column also has one of each.

R	B	G
G	R	B
B	?	?

The missing colors are: ___ and ___

4.2 — Complex Multi-Attribute Patterns

Each item has THREE attributes: shape, color, and size. Two attributes follow a pattern and one stays the same. Find what comes next.

Row 1: Shape changes, color changes, size stays



Next: Size=___ Color=___ Shape=___

Row 2: Size grows, shape stays, color changes



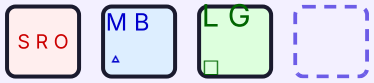
Next: Size=___ Color=___ Shape=___

Row 3: Color stays, shape changes, size grows



Next: Size=___ Color=___ Shape=___

Row 4: All three change each step



Next: Size=___ Color=___ Shape=___ (hint: the cycle resets)

S=Small, M=Medium, L=Large. R=Red, B=Blue, G=Green, P=Purple. O=Circle, ▵=Triangle, □=Square, ☆=Star

4.3 — Create Your Own Patterns

Now it is YOUR turn! Design your own patterns in the blank cells below. Follow the prompt for each row.

Make an AB pattern (two things alternating):



My rule: ___

Make an ABC pattern (three things repeating):



My rule: ___

Make a growing pattern (something increases each step):



My rule: ___

Make a shape pattern (use at least 2 different shapes):



My rule: ___

Make a number pattern (any rule you like!):



My rule: ___

4.4 — Pattern Puzzle Page

Mixed challenges! Each puzzle is different. Read the question carefully.

Puzzle 1 — What Comes Next?



Puzzle 2 — What Is Wrong? One item breaks the pattern. Circle it and write the correct one.



The wrong number is: ___. It should be: ___

Puzzle 3 — Fill in the Missing Piece.



Puzzle 4 — What Comes Next?



Hint: add the last two numbers.

Puzzle 5 — What Is Wrong? One shape breaks the mirror.



The wrong shape is in position: ___. It should be: ___

Puzzle 6 — Two Rules at Once. Fill in both blanks.



Next two: Color=___ Number=___ and Color=___ Number=___

Next Step

Open the NeuroKids main guide and pick your first weekly activity plan to keep the learning going!